January 7, 2025

To: Energy Division Rule 21 staff

Subject: **Informal Comments of the Vehicle-Grid Integration Council on 2024 Q4 Interconnection Discussion Forum Regarding "Outstanding Topics" and Forthcoming Rule 21 Proceeding Scope**

I. Introduction.

The Vehicle-Grid Integration Council ("VGIC") appreciates the opportunity to provide these informal comments on key topics related to the interconnection of vehicle-to-grid ("V2G") systems and associated processes under Rule 21. Below, we outline several areas where we believe improvements could significantly enhance the efficiency and clarity of interconnection procedures, ultimately supporting California's ambitious clean energy and transportation goals, including Senate Bill 676, which directs the Commission to maximize the use of feasible and cost-effective VGI.

II. Interconnection Application Form and Portal Updates to Allow Applicants to Indicate a Site as V2G.

To avoid confusion and delay, it is critical that each investor-owned utility's ("IOU") interconnection application forms and online portal, if available, allow applicants to clearly identify whether an energy storage system is a V2G system. This update, detailed although ultimately not directed in Decision (D.) 20-09-035 *Issue 23f*, is now more relevant as V2G interconnection requests have increased in the five years since the Working Group Three report was submitted. Furthermore, applicants should be able to specify the technology type (V2G AC or V2G DC), as it can expedite utility review and support the recommended reporting detailed in section VI below.

Notably, PG&E has made and maintained this update to its forms, which demonstrates that such updates are feasible and effective.[1] In contrast, SCE and SDG&E have not implemented similar updates, which VGIC members report resulting in confusion and delays.[2] The lack of appropriate forms also risks applications being routed out of Rule 21, for example, to a make-ready program or tariff (i.e., Rule 29/45).

[1] PG&E Electric Form No. 79-1174-03. Attachment H. Section AB. https://www.pge.com/tariffs/assets/pdf/tariffbook/ELEC_FORMS_79-1174-03H.pdf

[2] *See* SCE Electric Form 14-918. Question 3.b. https://www.sce.com/sites/default/files/custom-files/PDF_Files/ELECTRIC_FORM_14-918_PDF%20fillable.pdf and SDG&E Electric Form 142-05201 https://tariffsprd.sdge.com/view/tariff/?utilId=SDGE&bookId=ELEC&tarfKey=730

VGIC recommends this outstanding issue be scoped into the upcoming Rule 21 proceeding. Consistent, standardized processes are essential to enable V2G project development and avoid unnecessary delays.

III. Incorporate UL QIKP Certification Program as One Permanent Pathway for V2G AC Interconnection.

This item is referenced in issue #3 of *List of Potentially Outstanding or Important Interconnection Topics from Past IDFs, Proceedings, and Other Completed or Ongoing Interconnection Discussions.* VGIC recommends the upcoming Rule 21 proceeding seek to expedite the incorporation of the UL QIKP certification program for V2G AC systems as one available permanent interconnection pathway for V2G AC applicants. This could occur through a dedicated track in the Rule 21 proceeding, a dedicated reconvening of the V2G AC subgroup, or some other pathway. The UL QIKP certification pathway evaluates both the core UL 1741/UL 9741 safety requirements and the grid-interactive requirements found in UL 1741 SA and SB. **Timely incorporation into Rule 21 would immediately streamline interconnection for V2G AC projects,** which can significantly increase the availability of distributed V2G-capable systems supporting California's urgent ratepayer affordability, system reliability, and statewide climate goals.

Given these recent developments, as well as the highly self-contained nature of the issue relative to all other outstanding IDF/Rule 21 issues, VGIC strongly urges this issue be taken up in a near-term, dedicated effort to focus party efforts and streamline the incorporation of UL QIKP, rather than defer near-term action. **Critically, action to incorporate UL QIKP pathway should not be dependent on the completion of UL 1741 SC, which is a separate standard. VGIC sees no reason to wait until UL 1741 SC is published to consider the UL QIKP pathway, as these pathways are not mutually exclusive but rather complementary.**

Given the experience of VGIC, UL, utilities, and other participants in the recent Maryland docket incorporating a UL QIKP pathway for V2G AC interconnection (detailed below), we believe it is possible to efficiently address this issue in its own effort (either a dedicated track of the forthcoming Rule 21 proceeding, a dedicated reconvening of the V2G AC technical subgroup that is unrelated to the completion of UL 1741 SC, or some other dedicated effort). In turn, supporting V2G AC interconnection can unlock anywhere from 820 MWh[3] (i.e., only from currently-registered Tesla Cybertrucks in the state) to 150,000 MWhs[4] (from all on-road California EVs) of latent technical energy storage capacity. This untapped network of distributed energy storage customers could enroll in emergency reliability programs, cost-based export rate designs, and dynamic rate pilots which are intended to support the Commission's critical

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[3] Assumes an average 123 kWh Tesla Cybertruck battery size and 6,670 total Tesla Cybertrucks registered in California through Q3 2024. *See* California Energy Commission (2025). *New_ZEV_Sales_Last_update_11-19-2024_ada.xlsx.* ZEV and Infrastructure Stats Data. Data last updated September 30, 2024. Retrieved January 6, 2025 from https://www.energy.ca.gov/files/zev-and-infrastructure-stats-data.

[4] Assumes an average 71.4 kWh battery size and 2,113,135 EVs sold in California. See https://ev-database.org/cheatsheet/useable-battery-capacity-electric-car and https://www.veloz.org/ev-market-report/

affordability, reliability, and decarbonization goals. If even a very modest 0.25% of this total energy storage capacity from currently-registered Tesla Cybertrucks was made available for V2G activities, it would represent an immediate 400% increase in participating V2G kWh in California, thereby significantly supporting market development and ensuring widespread V2G can indeed support the grid in the long run, as envisioned by Senate Bill 676 (Bradford, 2019).[5] In terms of instantaneous available power, enabling the fleet of already-registered Tesla Cybertrucks to export to California's grid would represent 76 MW of instantaneous available export capability.[6] VGIC is concerned that the Commission may forego these significant values and the ratepayer benefits they bring if we defer consideration of the UL QIKP pathway until UL 1741 SC is complete.

As referenced above, there has been a recent regulatory precedent related to UL 1741 QIKP. Maryland's Public Service Commission recently indicated its approval of UL QIKP as one of two permanent interconnection pathway options for V2G AC systems, citing its strong technical foundation.[7] Specifically, the Commission documents note the following three reasons to offer the UL QIKP V2G AC pathway as an alternative to the UL 1741 SC/SAE J3072 approach:

1. "A core component of UL 1741 SC is that it enables the EVSE to authorize the EV to discharge based on the vehicle's compliance to SAE J3072. This is important in cases where several V2G capable vehicles may be plugging into the EVSE, and the EVSE needs to perform this check. However, in cases where only one V2G capable vehicle will be plugging into the EVSE (e.g., a single-family residential setting), the UL QIKP certification pathway is sufficient to ensure grid interconnection/performance certification. In summary, QIKP works for a single installation site with a single defined EV.
2. While the Workgroup supports the incorporation of UL 1741 SC as an option for V2G AC systems within these regulations, the UL 1741 SC working group has not yet published the standard and has pushed back publication a few times. Allowing for the QIKP pathway creates a hedge against further delays to UL 1741 SC, and enables Maryland to achieve the vision set forth in the DRIVE Act
3. Original Equipment Manufacturers (OEMs) of V2G systems may choose different pathways. As long as those pathways can ensure safety and reliability through UL

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[5] The two largest interconnected V2G sites in California are the Oakland and Cajon Valley school bus depots. The Oakland site uses 74 141-kWh BYD/RIDE Type A buses and the Cajon Valley uses six Lion Type C, which we will assume have the smallest battery capacity of 126 kWh. The available capacity from these sites totals 11 MWh. Assuming a 5% availability rate, this yields 0.5 MWh of available energy storage capacity. With 820 MWh available Tesla Cybetruck energy storage capacity at a 5% availability rate and a 5% customer participation rate, this represents a 367% increase.

[6] Assumes 11.5 kW V2G charging. https://www.tesla.com/powershare.

[7] *See:* Maryland Public Service Commission. *Notice Initiating a Rulemaking, Opportunity to Comment, and Rulemaking Session. RM 87.* https://webpscxb.psc.state.md.us/DMS/rm/rm87. *Proposed Regulations (RM87): PC44 Interconnection Workgroup: V2G Regulation Proposal.* October 30, 2024. Page 5-6. *See also* Commission approval of RM 87. https://www.youtube.com/watch?v=Ojh98h_81ZA&t=410s at 5:49. Note final adoption is not anticipated until May 1, 2025.

> certification, including both the fire/shock safety evaluation and the grid interconnection/performance functionality, then these pathways should be open to OEMs. This is particularly important while the V2G system market is still relatively nascent. We must ensure the core safety and reliability of the grid, which both pathways accomplish, while not being too prescriptive over the path that OEMs take."[8]

Utility and industry stakeholders have been briefed on the UL QIKP pathway for V2G AC systems, including at the December 7, 2023 Smart Inverer Working Group and subsequent meetings, and there exists growing support for this approach.

While early bidirectional charging products were based on a V2X DC configuration, more recent bidirectional charging solutions, including offerings from Tesla and Lucid, leverage a V2X AC configuration. Given the prevalence and continued growth of these systems, it is reasonable to address adoption of this pathway at this time.

Critically, adopting the UL QIKP pathway into Rule 21 via either (a) an immediate, separate track in the forthcoming Rule 21 proceeding, (b) an immediate reconvening of the V2G AC subgroup, or (c) some other pathway to immediately consider these revisions to Rule 21 *shall not* preclude progress on other likely viable V2G AC interconnection pathways standards, namely UL 1741 SC. Instead, it provides immediate optionality for developers while maintaining the core, rigorous safety and interoperability certification requirements needed to ensure safety and reliability on the grid.

With this in mind, VGIC strongly urges ED staff to consider the following scoping question for the new Rule 21 proceeding to address these deficiencies and resolve this important outstanding issue:

- "Should Rule 21 be updated in the immediate term, through either (a) a dedicated track of this proceeding, (b) a dedicated effort within the V2G AC subgroup, or (c) some other near-term approach to incorporate UL QIKP as a means of interconnecting V2G AC systems? Why or why not?
    - If so, what specific changes are needed to Rule 21 to allow this certification pathway to be utilized?"

Additionally, should UL 1741 SC be published prior to finalizing the new Rule 21 proceeding scope, and to ensure equitable treatment for both V2G AC standards pathways, the Commission may seek to pose the following related question to parties:

- "Should Rule 21 be updated in the immediate term, through either (a) a dedicated track of this proceeding, (b) a dedicated effort within the V2G AC subgroup, or (c) some other

[8] Maryland Public Service Commission. *Notice Initiating a Rulemaking, Opportunity to Comment, and Rulemaking Session. RM 87.* https://webpscxb.psc.state.md.us/DMS/rm/rm87. *Proposed Regulations (RM87): PC44 Interconnection Workgroup: V2G Regulation Proposal.* October 30, 2024. Page 5. Footnote 14.

near-term approach to incorporate UL 1741 SC as a means of interconnecting V2G AC systems? Why or why not?

- If so, what specific changes are needed to Rule 21 to allow this certification pathway to be utilized?"

IV. Reassess Application Fees for Non-NEM and Non-NBT Exporting Systems.

VGIC is concerned that interconnection fees for non-NEM and non-NBT exporting systems are disproportionately high and not reflective of actual utility costs. This is an outstanding issue from the previous Rule 21 proceeding R.17-07-007: "Should the Commission revise the application fee for non-net energy metering systems? How should the Commission determine what the revised fee should be?"[9] The topic is also detailed in issue #5 of *List of Potentially Outstanding or Important Interconnection Topics from Past IDFs, Proceedings, and Other Completed or Ongoing Interconnection Discussions.*

Notably, these fees significantly exceed those charged by utilities in other states, as demonstrated below in Attachment A. Additionally, these fees have an outsized impact on V2G systems, which have relatively lower upfront costs compared to stationary energy storage systems, rooftop solar PV, or other DERs interconnecting under Rule 21. For example, where the most significant share of the total upfront costs for a stationary battery energy storage system is the battery itself, this is not one of these costs in enabling a V2G system, as the vehicle battery has already been purchased for its primary transportation use case. As a result, the $800 interconnection application fee comprises one of the most significant upfront costs for V2G systems and, in turn, one of the largest barriers to widespread V2G.

We recommend this outstanding issue from R.17-07-007 be addressed in the first phase of the forthcoming Rule 21 proceeding to ensure the fees are reasonable for V2G customers. Reducing these fees would remove a significant barrier for smaller V2G projects, in particular.

V. Align Rule 21 with the CEC's DSGS Option 3.

The CEC's Demand Side Grid Support ("DSGS") Option 3 provides a critical revenue opportunity for V2G resources, with potential earnings of $100/kW-summer or more. However, the current UL 1741 SB requirement in Rule 21 prevents many V2G systems from participating, even though the CEC explicitly states UL 1741 SB listing is not required for DSGS Option 3 Battery Virtual Power Plants.

VGIC strongly urges this issue be resolved to unlock emergency reliability resources in response to the high-revenue V2G opportunity, as is the intention of CEC DSGS Option 3.

VI. Standardize Reporting of V2G Projects in Quarterly Rule 21 Timeline Filings.

[9] *Assigned Commissioner's Second Amended Scoping Memo and Ruling for Phase II of Proceeding.* May 12, 2021. http://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M384/K265/384265152.PDF. Page 10. Issue 9.

This outstanding item is related to the items detailed in issue #2 on the *List of Potentially Outstanding or Important Interconnection Topics from Past IDFs, Proceedings, and Other Completed or Ongoing Interconnection Discussions.* Decision 20-09-035 OP 22 mandates quarterly reporting on interconnection timelines, yet the lack of consistency in how V2G projects are categorized undermines the utility of these reports and the learnings used to inform broader program, policy, and tariff development at the CPUC.

PG&E's reporting does include detailed classifications, such as "Storage – EV" (presumably, V2G AC) and "Storage – EVSE" (presumably, V2G DC). This provides a model for transparency and supports key learnings to inform the development of CPUC initiatives. However, SCE and SDG&E quarterly timeline reporting fails to distinguish V2G technology types within the "energy storage" category, despite evidence of V2G interconnection activity in their territories (e.g., Cajon Valley School District in SDG&E territory). VGIC suspects that improvements made in response to the forms/portal update issue detailed above in section II would directly support this standardized reporting.

All IOUs should be required to uniformly categorize V2G projects in their reports. This will enable stakeholders to identify critical bottlenecks in the interconnection process and address them effectively, ensuring all technology types are treated equitably.

VII. Address Counterproductive Disconnect Requirements During Undervoltage Events.

This item is related to issues #3 and #7 in the *List of Potentially Outstanding or Important Interconnection Topics from Past IDFs, Proceedings, and Other Completed or Ongoing Interconnection Discussions.* As VGIC understands it, under current Rule 21 requirements, exporting devices must disconnect during undervoltage events. This approach is counterproductive, as it exacerbates grid instability rather than mitigating it. By staying connected, V2G systems could support local voltage levels and enhance grid resilience.

VGIC recommends revisiting this requirement to align with grid reliability objectives and enable V2G resources to contribute positively during undervoltage events.

VIII. Define a Timeline or Roadmap for Transitioning to Virtual Inspection and Notification-Only Approaches.

This item is related to issues #2 in the *List of Potentially Outstanding or Important Interconnection Topics from Past IDFs, Proceedings, and Other Completed or Ongoing Interconnection Discussions.* Virtual inspections and notification-only approaches both offer cost and time improvements over traditional site inspection processes, especially for smaller V2G projects. Establishing a clear timeline and milestones for this transition will streamline interconnection processes and reduce burdens on applicants.

Currently, all three utilities leverage virtual inspections to a certain extent to streamline interconnection. However, field inspections are still required typically for projects using relays, AC disconnects, NGOMs, and variances from Rule 21. Meanwhile, the currently-approved notification-only pilot is available through September 2025, and is only open to non-export standalone storage or solar-plus-storage projects less than 30 kVA capacity.[10]

VGIC acknowledges that the V2G market may not yet be mature enough to leverage these approaches. However, we recommend the new scope of the proceeding consider establishing certain milestones or explicit timelines for when both issues should be taken up. Presumably, virtual inspections would be applied to V2G in the near term, whereas notification-only may be a more mid- to long-term endeavor. VGIC advises against subjecting these issues to the organic cycles of scoping new Rule 21 issues – as evident in Phase II of R.17-07-007, unexpected events can lead to delays of 4-5 years, even for already scoped, time-sensitive matters. Instead, VGIC recommends the new proceeding consider what milestones (e.g., a certain number of deployments) or timelines (e.g., a certain date for implementation) are appropriate to implement virtual inspection and, subsequently, notification-only approaches.

IX. Conclusion

VGIC appreciates the opportunity to submit these informal comments to the CPUC Energy Division staff on outstanding or important interconnection topics. We look forward to further collaboration with ED staff and other stakeholders on this important initiative.

Respectfully submitted,

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[10] https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M528/K052/528052750.PDF